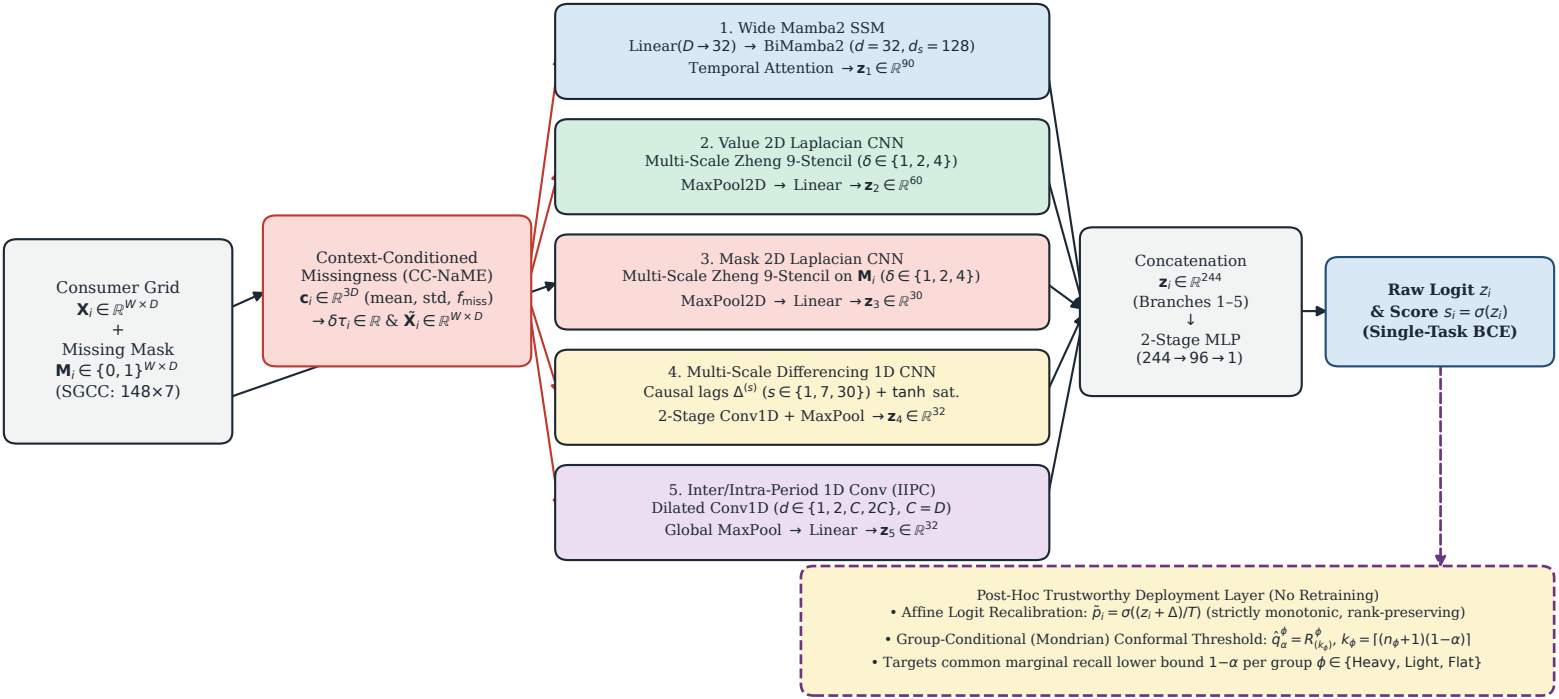

MAMRD (a34_generic): End-to-End Multi-Branch Detector + Post-Hoc Conformal Audit
Consumer Grid
$\mathbf{X}_i \in \mathbb{R}^{W \times D}$
+
Missing Mask
$\mathbf{M}_i \in \{0,1\}^{W \times D}$
(SGCC: 148×7)
Context-Conditioned
Missingness (CC-NaME)
$\mathbf{c}_i \in \mathbb{R}^{3D}$ (mean, std, f_{miss})
$\rightarrow \delta\tau_i \in \mathbb{R}$ & $\tilde{\mathbf{X}}_i \in \mathbb{R}^{W \times D}$
1. Wide Mamba2 SSM
Linear($D \rightarrow 32$) → BiMamba2 ($d = 32, d_s = 128$)
Temporal Attention → $\mathbf{z}_1 \in \mathbb{R}^{90}$
2. Value 2D Laplacian CNN
Multi-Scale Zheng 9-Stencil ($\delta \in \{1, 2, 4\}$)
MaxPool2D → Linear → $\mathbf{z}_2 \in \mathbb{R}^{60}$
3. Mask 2D Laplacian CNN
Multi-Scale Zheng 9-Stencil on $\mathbf{M}_i$ ($\delta \in \{1, 2, 4\}$)
MaxPool2D → Linear → $\mathbf{z}_3 \in \mathbb{R}^{30}$
4. Multi-Scale Differencing 1D CNN
Causal lags $\Delta^{(s)}$ ($s \in \{1, 7, 30\}$) + tanh sat.
2-Stage Conv1D + MaxPool → $\mathbf{z}_4 \in \mathbb{R}^{32}$
5. Inter/Intra-Period 1D Conv (IIPC)
Dilated Conv1D ($d \in \{1, 2, C, 2C\}$, $C = D$)
Global MaxPool → Linear → $\mathbf{z}_5 \in \mathbb{R}^{32}$
Concatenation
$\mathbf{z}_i \in \mathbb{R}^{244}$
(Branches 1–5)
↓
2-Stage MLP
($244 \rightarrow 96 \rightarrow 1$)
Raw Logit z_i
& Score $s_i = \sigma(z_i)$
(Single-Task BCE)
Post-Hoc Trustworthy Deployment Layer (No Retraining)
• Affine Logit Recalibration: $\tilde{p}_i = \sigma((z_i + \Delta)/T)$ (strictly monotonic, rank-preserving)
• Group-Conditional (Mondrian) Conformal Threshold: $\hat{q}_\alpha^\phi = R_{(k_\phi)}^\phi$, $k_\phi = \lceil (n_\phi + 1)(1-\alpha) \rceil$
• Targets common marginal recall lower bound $1-\alpha$ per group $\phi \in$ {Heavy, Light, Flat}